Individual Contribution Report

1. Introduction

In this project, we developed deep learning models to classify facial images into four pain intensity levels: no pain, mild, moderate, and severe.

My contribution focused on implementing and analyzing two models:

- A Custom CNN as a baseline model
- An advanced EfficientNet-B3 model using transfer learning

The goal was to compare their performance and improve pain classification accuracy.

2. Dataset Description

The dataset consists of RGB facial images mapped into four pain categories:

- No Pain
- Mild Pain
- Moderate Pain
- Severe Pain

Although the dataset is balanced, the classification task remains challenging due to:

- Subtle differences between classes
- Variations in lighting, pose, and facial orientation
- Noise introduced by emotion-to-pain label mapping

3. My Contribution

My main contributions include:

- Implementing a Custom CNN baseline model
- Implementing EfficientNet-B3 using transfer learning
- Designing a two-phase training strategy
- Applying class weighting to handle imbalance
- Evaluating models using:
    - Accuracy
    - F1-score
    - Confusion matrix
- Performing comparative analysis between models

4. Custom CNN Model (Baseline)

Model Description

The Custom CNN was designed as a baseline model and trained from scratch. The Custom CNN consists of:

- Multiple convolutional layers (Convolutional layers for feature extraction)
- Activation functions (ReLU)

- Pooling layers
- Fully connected classifier

This model was trained **from scratch** without pretrained weights.

Results

- Accuracy: **28.35%**
- F1 Score: **26.14%**

Analysis

- The model shows **poor performance**
- High confusion between all classes
- Strong bias toward predicting *no pain*

Training Behavior

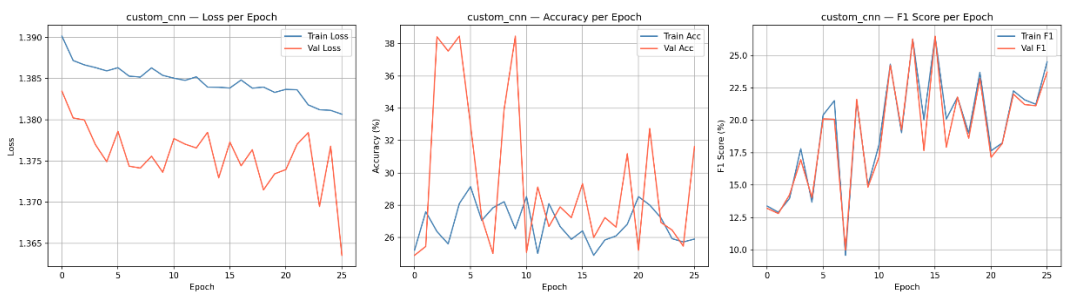

Figure 1: Training Curves for Custom CNN

The training curves show slow improvement with low accuracy and unstable validation performance.

Confusion Matrix

The confusion matrix shows significant misclassification across all classes, with a strong bias toward predicting *no pain*.

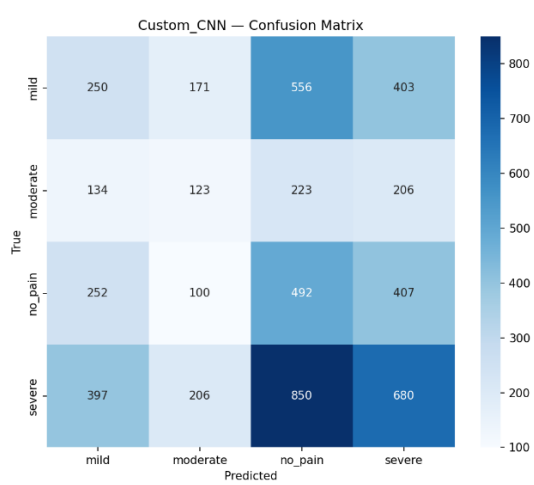

Figure 2: Confusion Matrix for Custom CNN

Conclusion:

The Custom CNN demonstrates underfitting, indicating that it is unable to learn meaningful facial features. This highlights the limitation of training from scratch on this dataset.

5. EfficientNet-B3 Model

Model Description

EfficientNet-B3 is a pretrained deep neural network optimized for image classification.

Modifications include:

- Replaced final layer for 4-class classification
- Added dropout to reduce overfitting

Training Strategy

A two-phase training approach was used:

Phase 1 – Feature Extraction

- Backbone frozen
- Only classifier trained

Phase 2 – Fine-Tuning

- Entire network unfrozen (All layers)
- Trained with lower learning rate

Results

- Accuracy: 78.11%
- F1 Score: 77.64%
- Precision: 77.30%
- Recall: 78.05%

Class-wise Performance

- Severe pain → highest accuracy
- Mild pain → lowest accuracy

Indicates:

- Strong expressions are easier to detect
- Subtle expressions are more challenging

Training Behavior

The model shows strong convergence, with high training accuracy (~97%) and stable validation performance (~78%).

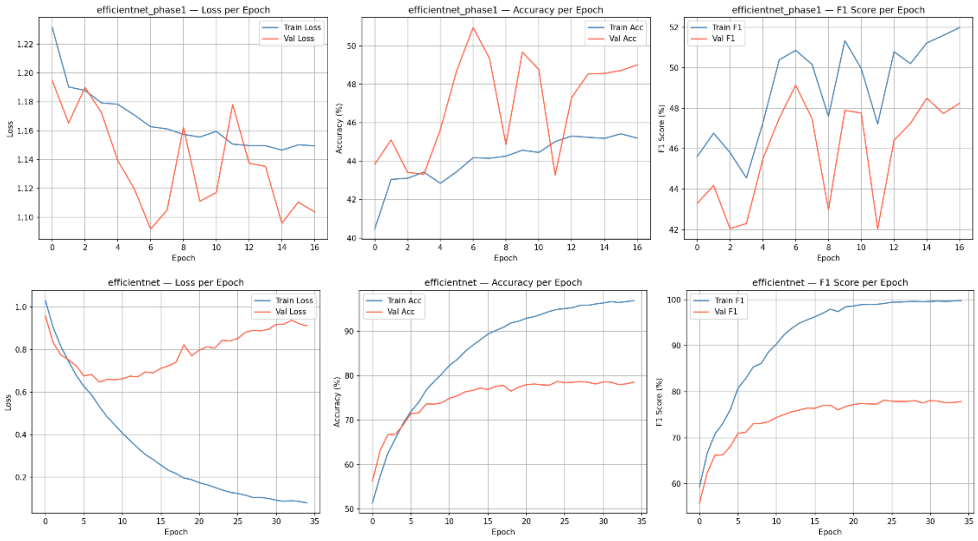

Figure 3: Training Curves for EfficientNet-B3

Confusion Matrix

The confusion matrix shows strong diagonal dominance, indicating accurate predictions across most classes.

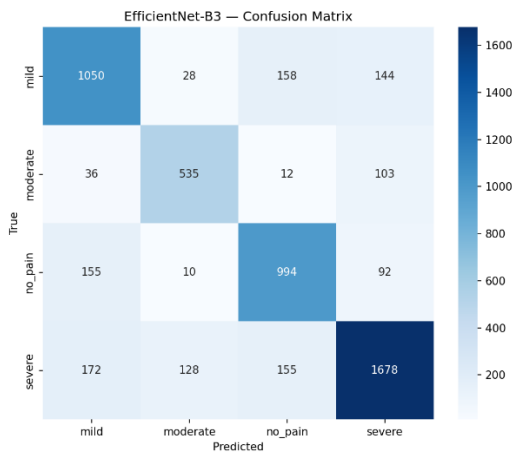

Figure 4: Confusion Matrix for EfficientNet-B3

Analysis:

EfficientNet significantly outperforms the baseline model. It successfully captures complex facial patterns and achieves strong generalization.

6. Model Comparison

Model	Accuracy	F1 Score
Custom CNN	28.35%	26.14%
EfficientNet	78.11%	77.64%

EfficientNet improves performance by **more than 50% in F1-score**

7. Analysis

- EfficientNet shows strong learning and stable convergence

- Custom CNN underfits and fails to extract useful features
- Remaining errors occur mainly between:
    - Mild vs Moderate
    - Moderate vs No Pain

Important insight:

Incorporating localized facial information provides additional discriminative power, particularly for distinguishing between subtle pain levels such as mild and moderate.

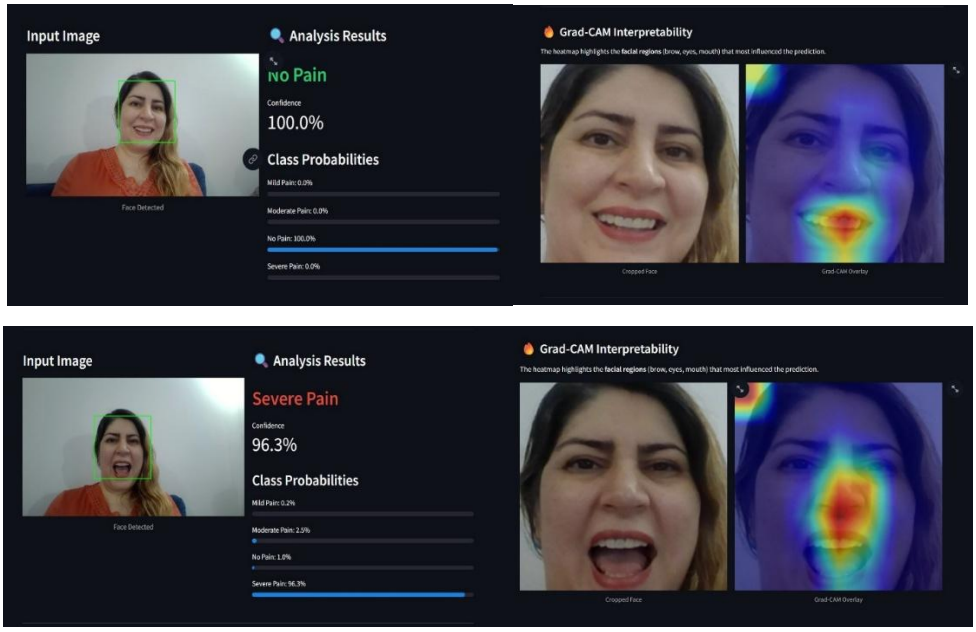

Figure 5: Example predictions and Grad-CAM visualizations from EfficientNet model

8. Challenges

- Subtle differences between pain levels
- Dataset labels derived from emotions (noise)
- Class imbalance

9. Conclusion

- Transfer learning significantly improves performance
- EfficientNet-B3 is highly effective for this task
- Two-phase training is critical for achieving strong results

Future improvements:

- Multi-input models
- More accurate clinical datasets

IX. Percentage of Code Contribution

Approximately 65% of the Streamlit user interface was adapted from publicly available online resources. The remaining 35% includes model implementation, training pipeline design, and project-specific modifications, which were developed and customized as part of my individual contribution to this project.